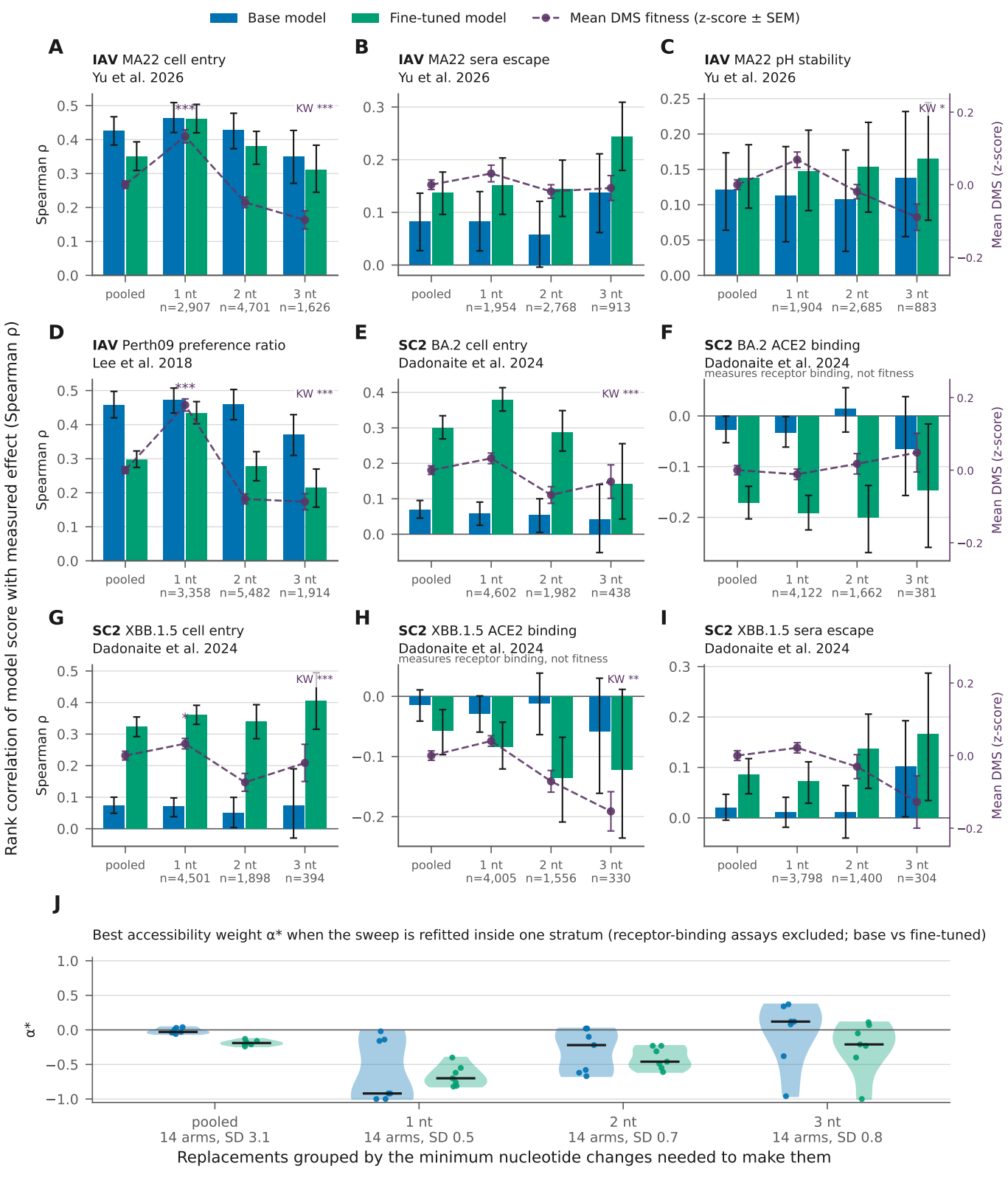

Base model
Fine-tuned model
Mean DMS fitness (z-score ± SEM)
A
IAV MA22 cell entry
Yu et al. 2026
B
IAV MA22 sera escape
Yu et al. 2026
C
IAV MA22 pH stability
Yu et al. 2026
D
IAV Perth09 preference ratio
Lee et al. 2018
E
SC2 BA.2 cell entry
Dadonaite et al. 2024
F
SC2 BA.2 ACE2 binding
Dadonaite et al. 2024
measures receptor binding, not fitness
G
SC2 XBB.1.5 cell entry
Dadonaite et al. 2024
H
SC2 XBB.1.5 ACE2 binding
Dadonaite et al. 2024
measures receptor binding, not fitness
I
SC2 XBB.1.5 sera escape
Dadonaite et al. 2024
Rank correlation of model score with measured effect (Spearman ρ)
Spearman ρ
Mean DMS (z-score)
pooled
1 nt
2 nt
3 nt
KW ***
KW **
KW *
J
Best accessibility weight α* when the sweep is refitted inside one stratum (receptor-binding assays excluded; base vs fine-tuned)
α*
pooled
14 arms, SD 3.1
1 nt
14 arms, SD 0.5
2 nt
14 arms, SD 0.7
3 nt
14 arms, SD 0.8
Replacements grouped by the minimum nucleotide changes needed to make them